IBM Z DATATHON 2026 · TRACK: REAL-TIME AI FOR CRITICAL DECISIONS

Real-Time Transfer Scam Guard

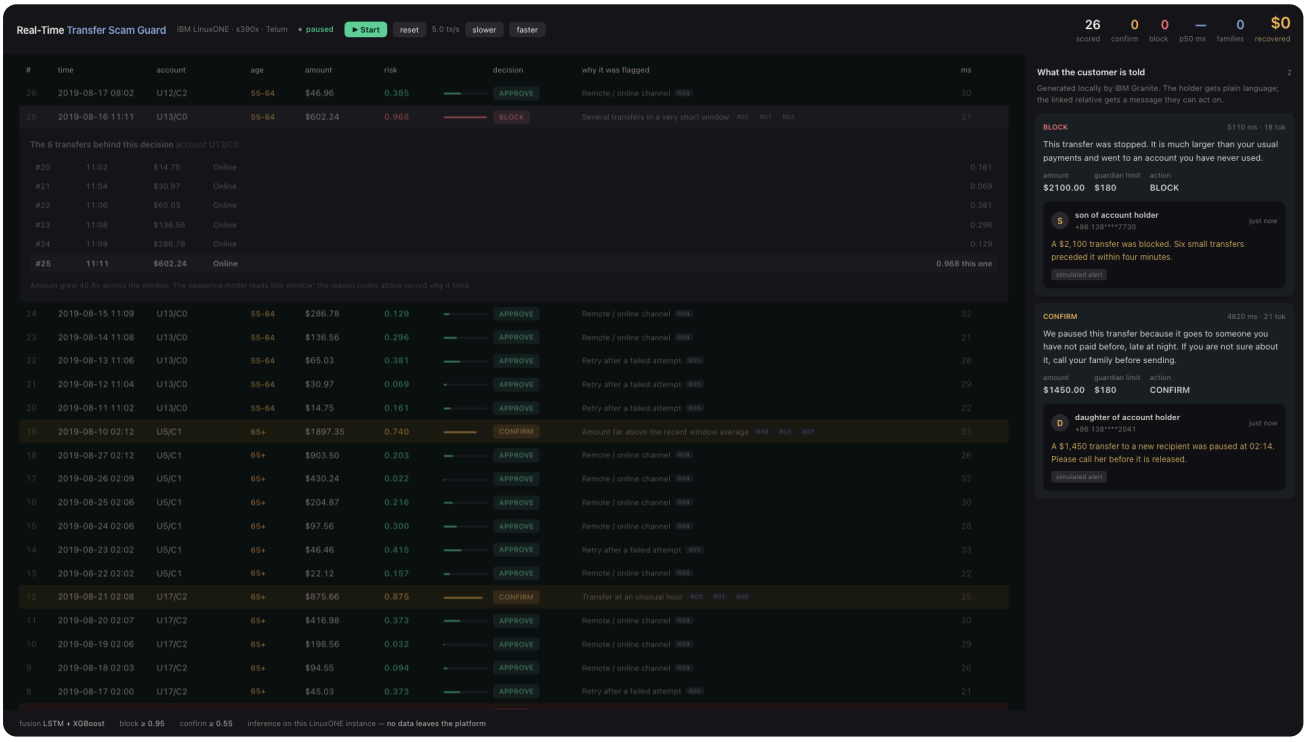
Scores every transfer before the money leaves, and returns APPROVE, CONFIRM or BLOCK in about 22 ms.

65.2%
FRAUD RECALL

96.7%
PRECISION ON STOPS

22 ms
MEDIAN DECISION

$18,418
RECOVERED



The console, running on IBM LinuxONE (s390x). Each row is one decision. The open panel shows the six transfers behind a blocked one: same account, a few minutes apart, amounts climbing from $14.75 to $602.24.

What it is

A decision engine that sits inside the bank's authorisation path. When the risk is elevated it pauses the transfer, explains it to the account holder in plain language, and alerts a linked family member.

Why it matters

Older people authorise scam transfers themselves, so nothing looks wrong until the money clears. The few seconds before authorisation are the only place to intervene.

SECTION 1 · THE PROBLEM

Older people send the money themselves

To the bank the payment looks normal: the customer authenticated properly, from a device they have used before. Nothing is wrong until the money clears, and by then it is gone. The intervention point is the few seconds before authorisation.

Victim tricked	Transfer authorised	Funds cleared	Bank / family notice
Hours of phone pressure	Valid PIN, valid device	Irreversible	Too late

$4.885 billion reported lost by older Americans in 2024, across 147,127 complaints, 43% more than 2023.

The FBI notes reporting is incomplete, because victims are embarrassed or do not know they were scammed. Source: FBI IC3, 2025. Existing fraud systems detect misuse after the fact; they were not built for a customer being talked into paying right now, and they cannot involve the family.

SECTION 2 · THE SOLUTION

Three outcomes, decided in the authorisation path

APPROVE

Release immediately

Risk below 0.55. Normal payments are not slowed down.

CONFIRM

Pause, explain, alert the family

Risk 0.55 to 0.95. The holder gets plain language. Above their guardian limit, a linked relative is alerted too.

BLOCK

Hold and record

Risk above 0.95. The transfer is held and an auditable record is written with reason codes.

Bringing in the family

A blocked card can be retried at another branch or online. A relative who knows the customer can end the call. CONFIRM is the outcome that pauses the payment and brings that person in.

Auditable by default

Nine reason codes (R01 to R09) travel with every decision, so a case can be reviewed later and the model audited over time. Clicking any row in the console opens the transfers behind that decision.

SECTION 3 · RESULTS

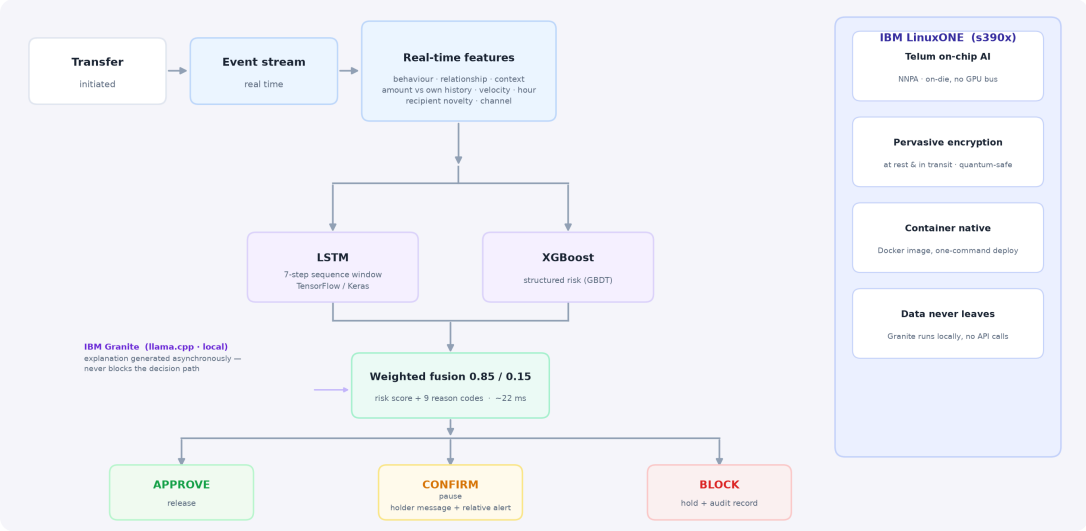
Measured on the LinuxONE instance

65.2%	96.7%	22 ms	$18,418
FRAUD RECALL	PRECISION ON STOPS	MEDIAN DECISION	RECOVERED

OFFLINE QUALITY, HELD-OUT TEST SPLIT (0.11% POSITIVES)	PR-AUC	ROC-AUC	BEST F1
LSTM only	0.3829	0.9293	0.478
XGBoost only	0.0765	0.9411	0.161
Fusion (0.85 / 0.15)	**0.4277**	**0.9654**	**0.506**

SECTION 4 · ARCHITECTURE

Two models, one decision, about 22 ms



Only the decision sits in front of the customer. IBM Granite writes the explanation afterwards and never delays authorisation.

Why two models

A 7-step LSTM reads the sequence of recent transfers; an XGBoost GBDT reads the structured risk. The scores are fused 0.85 / 0.15. XGBoost alone catches few frauds at usable precision, and the LSTM alone misses structured signals such as amount against the account's own history.

Why it stays fast

Feature construction and scoring are numeric work on the CPU. The language model runs on two threads in a separate pool, so writing an explanation cannot delay a decision.

SECTION 5 · PLATFORM

Running on IBM Z / LinuxONE

- **Telum on-chip AI.** Both models and Granite run on the same processor; the accelerator shares the die with the CPU.
- **Pervasive encryption.** Data is encrypted at rest and in transit by the platform, quantum-safe.
- **Data stays on platform.** Granite is served locally by llama.cpp with the zDNN / NNPA backend. No external API calls.
- **Container native.** One Docker image on IBM's s390x AI/ML base, health-checked. `uname -m` returns `s390x`.

SECTION 6 · SCOPE

What it does and does not do

- Scores each transfer in about 22 ms and returns one of three decisions
- Writes a plain-language message for the holder, with reason codes
- Alerts a linked relative above the guardian limit
- Runs containerised on LinuxONE, with no API calls
- No live bank core integration; the stream is a replay
- The guardian notification is simulated, not a real SMS gateway
- No production load testing; age fields are a synthetic overlay
- onnxruntime has no s390x wheel, so ONNX is export-only

Next 30 days: swap the replay for the bank's authorisation feed, connect a real notification provider, and run IEEE-CIS and PaySim as the production corpus. Code, benchmark and tests: github.com/anvictor042-cyber/real-time-transfer-scam-guard (MIT).